Critical Path Analysis

The Critical Path Method (CPM) is a project management technique used for planning, scheduling, and controlling projects. It helps determine the minimum time required to complete a project and identifies activities that directly impact the project completion date. CPM is especially useful for complex projects with multiple interdependent tasks.

- A critical task is an activity that cannot be delayed without delaying the overall project completion. Any delay in a critical task immediately affects the project deadline.
- The critical path is the longest sequence of dependent activities in a project network. It represents the shortest possible project duration. All activities on this path are called critical activities, and they have zero float.

Relationship Between Critical Task and Critical Path

A critical task (activity) and the critical path are closely related concepts in the Critical Path Method:

- A critical task is any activity with zero float, meaning it cannot be delayed without delaying the project.
- The critical path is the continuous sequence of critical tasks from the start of the project to the end.
- Every activity on the critical path is a critical task, but not every project activity is critical.

- Delaying any single critical task will delay the entire critical path, and therefore delay the project completion.

Benefits of Using Critical Path Method

- Provides a clear visual representation of the project schedule
- Identifies the most important (critical) tasks
- Helps anticipate and manage project risks
- Improves communication and coordination within the project team

Steps to Find the Critical Path

1. Identify all activities required to complete the project
2. Define the sequence and dependencies of activities
3. Estimate the duration of each activity
4. Draw the Activity-on-Node (AON) network diagram
5. Perform a forward pass to calculate earliest times
6. Perform a backward pass to calculate latest times
7. Calculate float and identify the critical path

Project Activity Data

The table given below contains the activity label, its respective duration (in weeks), and its precedents. We will use the critical path method to find the critical path and activities of this project.

Activity	Duration (in weeks)	Precedents
A	6	-
B	4	-
C	3	A
D	4	B
E	3	B
F	10	-
G	3	E,F
H	2	C,D

Rules for Activity-on-Node (AON) Network Diagram

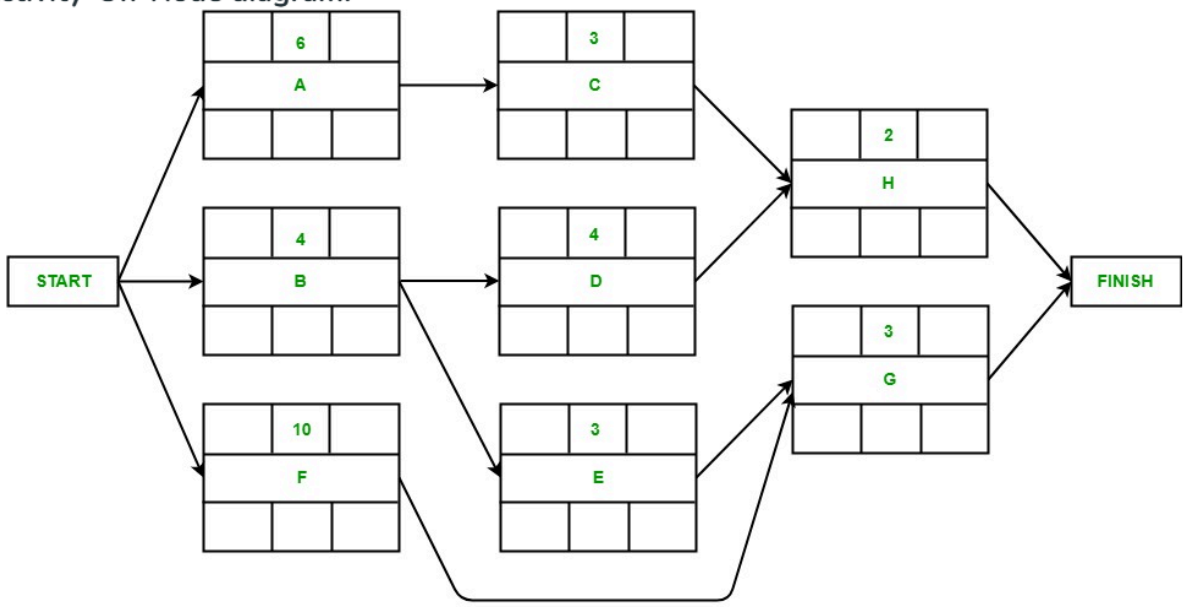
- Only one start node and one end node
- Each node represents an activity and has a duration
- Arrows show precedence relationships and have no duration
- Time flows from left to right
- The network should not contain loops or dangling activities

Node Representation:

Earliest Start	Duration	Earliest Finish
Activity Label		
Latest Start	Float	Latest Finish

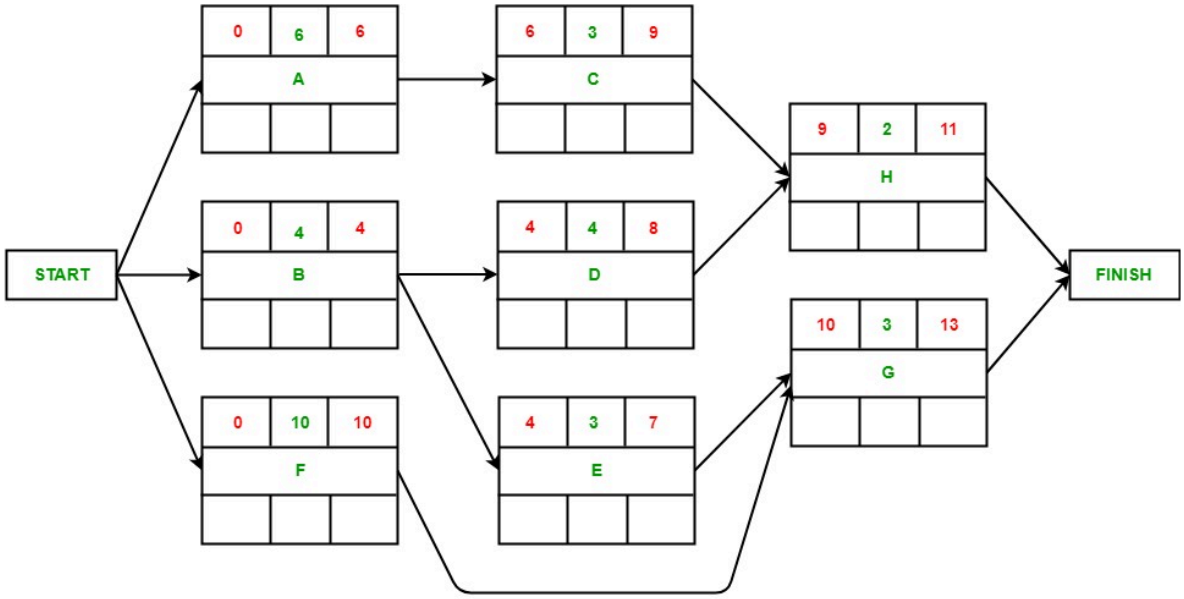
- **ES (Earliest Start):** Earliest time an activity can begin
- **EF (Earliest Finish):** ES + Duration
- **LS (Latest Start):** Latest time an activity can begin without delay
- **LF (Latest Finish):** LS + Duration
- **Float:** LS – ES or LF – EF

Activity-On-Node diagram:



Forward Pass (Earliest Times Calculation)

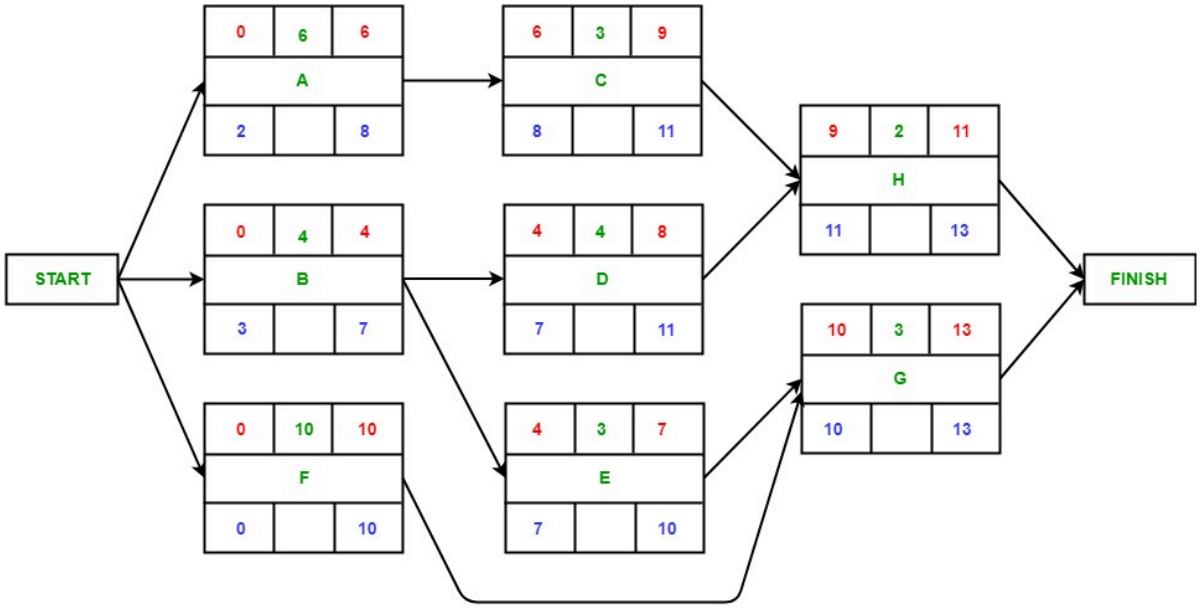
The forward pass is carried out to calculate the earliest dates on which each activity may be started and completed.



Backward Pass (Latest Times Calculation)

The backward pass is carried out to calculate the latest dates on which each activity may be started and finished without delaying the end date of the project.

Assumption: Latest finish date = Earliest Finish date (of project).



Backward Pass in Critical path in project management

Identifying the Critical Path

The critical path is identified by calculating the float (slack) for each activity. Activity Float represents the amount of time an activity can be delayed without affecting the overall project completion date. It is calculated using either of the following formulas:

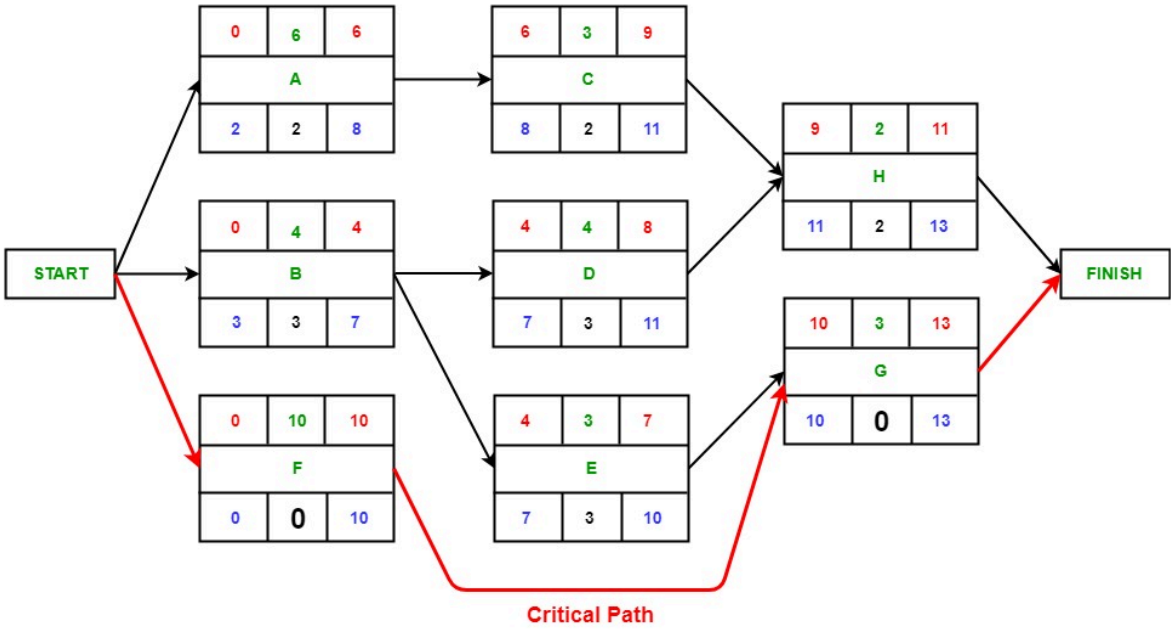
Float = LS − ES
or
Float = LF − EF

An activity with **zero float** is a **critical activity**, meaning any delay in that activity will directly delay the completion of the entire project.
Activity float is calculated as:

- **Float = LS − ES = LF − EF**

Activities with **zero float** are critical activities. In this project:

- **Activity F:** Float = 0
- **Activity G:** Float = 0

Critical Path

Start → F → G → End